

1 (a) force per unit mass (*ratio idea essential*) B1 [1]

(b) $g = GM / R^2$ C1
 $8.6 \times (0.6 \times 10^7)^2 = M \times 6.67 \times 10^{-11}$ C1
 $M = 4.6 \times 10^{24} \text{ kg}$ A1 [3]

(c) (i) *either* potential decreases as distance from planet decreases
or potential zero at infinity and X is closer to zero
or potential $\propto -1/r$ and Y more negative M1
 so point Y is closer to planet. A1 [2]

(ii) idea of $\Delta\phi = \frac{1}{2}v^2$ C1
 $(6.8 - 5.3) \times 10^7 = \frac{1}{2}v^2$
 $v = 5.5 \times 10^3 \text{ ms}^{-1}$ A1 [2]

2 (a) *either* the half-life of the source is very long